

Product Evaluation — Live Translate

- **Student:** Eva Losada
- **Date:** 2026-07-14
- **Video demo:** demo.mp4 — full-page on-page walkthrough on theblogstarter.com/about-me
- **LLM provider / model:** Anthropic Claude — claude-sonnet-5
- **Backend target:** rubric + benchmark run against local (http://localhost:8787 gateway → private AI service :8000); **public deploy verified live** at https://fde-live-translate-gateway.fly.dev (auto-checked, deploy_health_ok: true).

Verdict

This is a genuinely shippable Live Translate backend with a production-shaped topology: a public Node gateway is the only internet-facing surface, and the Python AI service, API key, model, and two-tier cache sit behind it privately. Run end-to-end against real third-party content, it performs well — natural **Mexican Spanish** (idiomatic recámara , plomero , Haz ejercicio), every number/price/unit preserved, a genuine cold-cache benchmark showing a **322x hit/miss speedup** and **~\$59.79/mo saved** at 500k requests, 0 errors, all SLAs met, and request-ID tracing that is greppable end-to-end across both services. Both services are **deployed and live on Fly.io** — the public gateway /health nests the private AI service, and a real translate through the public URL preserves numbers and prices in es-MX. The machines auto-stop when idle and auto-wake on first request (~3-5s cold start), so the deploy runs cost-efficiently while staying reachable. Submission-ready across every dimension.

Rubric score (from eval/report.json): 70 / 70 auto (+ 30 manual)

1. Performance & cost (from benchmark/bench.py , genuine cold-cache run)

Metric	Result	SLA	Pass?
Cache hit p95	10.5 ms	≤ 60 ms	✓
Cache miss p95	3375.7 ms	≤ 3500 ms	✓
Cache hit rate	75.0 %	≥ 60 %	✓
Throughput	1499.8 req/s	≥ 20	✓
Error rate	0.0 %	≤ 1 %	✓
Cost per miss	\$0.0001594	—	—
Monthly savings from cache	\$59.79	—	—

python benchmark/bench.py exits 0 — **all SLAs met**. Numbers are from a genuine cold run against an **empty** cache on 2026-07-14 (miss p95 3.4s is a real LLM call; hit p95 10.5ms is served from the two-tier cache → **322x** speedup). Cost is modeled at 500k req/mo: **\$79.72 uncached** → **\$19.93 cached**. Raw output in benchmark/_bench.json .

2. Live-website test

The primary demonstration is the **on-page video demo** (demo.mp4) on a real third-party site the student does not control. So the translations are independently checkable in this document, the same page's strings were also re-run live through the public gateway (raw request/response saved to eval/aboutme_live_test.json).

- **On-page video demo** — `theblogstarter.com/about-me` : `demo.mp4` loads the widget via the Chrome extension (pointed at the public Fly.io gateway) and does **full-page one-click Translate** → **Restore** on this live page. The whole page — heading, nav, body, sidebar, and comments — flips to Mexican Spanish; the on-screen badge reports **67 chunks · 8 cache hits · ~16.8 s**. This is the primary live, on-page demonstration.
- **Same page verified through the live gateway**: 8 representative strings from `theblogstarter.com/about-me` were sent through the live public gateway (`POST /translate/batch` , `target: es-MX`) on 2026-07-14; **all 8 returned correct es-MX** (table below), server `latencyMs 3964` on the uncached strings.
- **Cache on re-translate**: re-sending the identical strings returned **all 8 cached: true** with server `latencyMs 0` — the second call did zero LLM work, matching the cache-hit badge shown in the video.
- **Coverage**: Full-page coverage (heading, nav, body, sidebar) is shown on-screen in the video; no gaps or swallowed English. On strict-CSP sites the Chrome extension is required for in-page translation (console injection is blocked by CSP — a browser limitation, not a backend fault).
- **Resilience**: Clean — 0 errors, layout intact, correct response shape, no untranslated fallback. The widget was hardened this session to resolve the backend URL lazily (avoids a config race when injected as a content script).

Sample translations (live `theblogstarter.com/about-me` content)

Original (EN)	Translation (es-MX)	Numbers/symbols kept?	OK?
My name is Scott Chow, and I wrote the easiest guide to starting a blog, so that you can start your blog today!	Me llamo Scott Chow, y escribí la guía más fácil para empezar un blog, ¡para que puedas comenzar tu blog hoy mismo!	—	✓
Technology keeps changing at a rapid pace.	La tecnología sigue cambiando a un ritmo acelerado.	—	✓
When thinking about what to call the website, I decided that my nickname would work great.	Cuando estaba pensando en cómo llamarle al sitio web, decidí que mi apodo quedaría muy bien.	—	✓
If you want to see more from me, follow me on Facebook, Twitter and Pinterest.	Si quieres ver más contenido mío, sígueme en Facebook, Twitter y Pinterest.	—	✓
Recent comments from The Blog Starter users	Comentarios recientes de los usuarios de The Blog Starter	—	✓
Helping start blogs since 2002.	Ayudando a iniciar blogs desde 2002.	✓ 2002	✓
Steps for building your blog	Pasos para construir tu blog	—	✓
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Register is unmistakably **Mexican** (`Me llamo` , informal `puedas / quieres / sígueme` , `apodo`), translation-only with no preamble or wrapping quotes.

Price / number / unit preservation (verified live through the public gateway):

Original (EN)	Translation (es-MX)	Kept verbatim
Supplements cost about \$47 per month.	Los suplementos cuestan alrededor de \$47 al mes.	✓ \$47
The premium plan is \$12.99/month — save 20% with the annual plan.	El plan premium cuesta \$12.99/mes — ahorra 20% con el plan anual.	✓ \$12.99 · 20%
Package weight: 2.5 kg (5.5 lb), dimensions 30×20×10 cm.	Peso del paquete: 2.5 kg (5.5 lb), dimensiones 30×20×10 cm.	✓ 2.5 kg · 5.5 lb · 30×20×10 cm

Also confirmed on the deploy with `$129.99` (see the Deployment row below).

3. Dimension scorecard

Dimension	Pass / Partial / Fail	Evidence
Translation accuracy	Pass	8/8 live <code>theblogstarter.com/about-me</code> strings fluent and faithful; full-page translate shown in the video.
Mexican-Spanish register (es-MX)	Pass	Me <code>llamo</code> , informal <code>puedas / quieres / sígueme</code> , <code>apodo</code> — not Castilian/generic.
Numbers / symbols preserved	Pass	<code>2002</code> , <code>© 2026</code> (about-me page) plus <code>\$47</code> and <code>\$129.99</code> (live gateway) all preserved verbatim.
Page coverage	Pass	Full-page translate/restore on <code>theblogstarter.com/about-me</code> shown on-screen in the video (67 chunks: heading, nav, body, sidebar, comments); widget walks the whole page via <code>createTreeWalker(document.body, SHOW_TEXT)</code> (<code>translation-widget.js</code>).
Cache effectiveness	Pass	322× cold benchmark (10.5 ms hit p95 vs 3375.7 ms miss p95); video shows the cache-hit badge, and the live re-run of the same page returned all <code>cached: true</code> at <code>latencyMs 0</code> ; SQLite survives restart.
Latency vs SLA	Pass	Every SLA in <code>sla.json</code> met; <code>bench.py</code> exits 0.
Error handling (no silent English)	Pass	<code>lib/llm.py</code> fails loud (no untranslated-fallback); bad input → 400 (auto-verified).
Resilience on a real site	Pass	0 errors on live content; extension injects on strict-CSP sites; widget URL-race fixed.
UX polish	Pass	One-click translate/restore, cache-hit badge; backend hint now refreshes on panel open.
Deployment (Fly.io public URL)	Pass	Both services live on Fly.io; public gateway <code>/health</code> nests AI; live translate via public URL → <code>El taladro sin cable cuesta \$129.99 en el pasillo 7.</code> (\$129.99 preserved, es-MX). <code>deploy_health_ok: true</code> .

Also verified this session: **trace correlation** — an injected `X-Request-Id` (`eval-trace-4576aa`) appeared in **both** the gateway log (`POST /translate 200 1941ms`) and the AI-service log (`request_id`), greppable end-to-end. No secrets, `*.db` , or `*.log` are tracked in git.

4. Top fixes before shipping

No blocking gaps — all dimensions Pass. Minor notes:

1. **First-request cold start (~3-5s)**. Fly machines auto-stop when idle to save cost; the first translate after a quiet period cold-starts them. Acceptable for grading; if a warm demo matters, set `min_machines_running: 1` on the gateway.
2. **Extension backend URL**. The extension's saved backend URL is set to the public gateway (`https://fde-live-translate-gateway.fly.dev`) — confirm this before any future demo so the on-page test exercises the deploy rather than localhost.